

# The potential for an Oakland Green Bank and Green Opportunities Fund

March 4, 2024

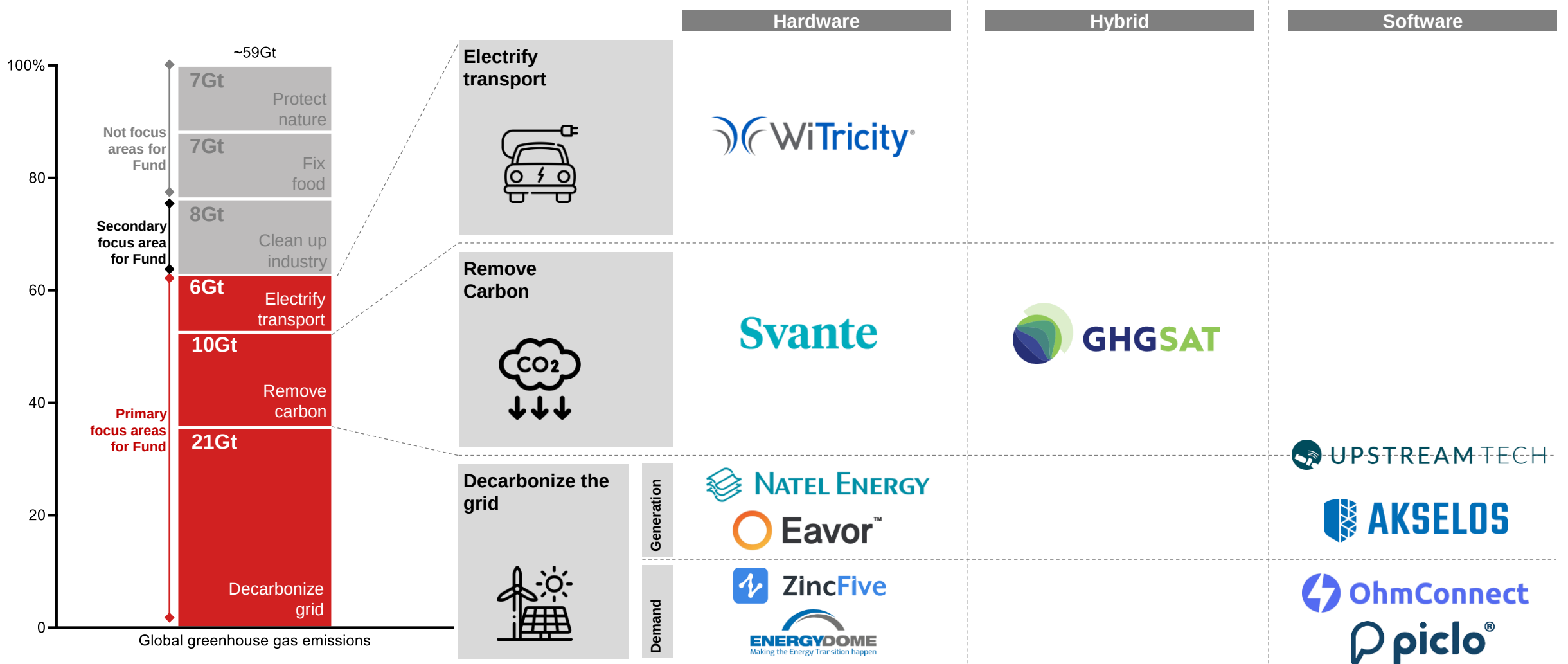


# Agenda

- Introduction
- Context and Oakland ECAP
- Green Bank
- Green Opportunities Fund
- Q&A / Discussion

Japan Energy Fund is a growth stage climate-tech venture capital fund backed by Japanese corporations and investing in startups globally

## Investment areas and portfolio companies



Oakland faces historical and present challenges but also has the opportunity to ‘get richer by becoming greener’

## Challenges

- **Quality of life:** social inequality / extraction; public transport & infrastructure; disruption from historical developments & politics
- **Local investment:** lower economic power compared to adjacent areas e.g., Berkeley, San Francisco
- **Climate change:** mitigation & adaptation needs
- **Financing:** access for higher-risk, novel business models



## Vision



Equitable, sustainable revitalization



Local, homegrown economic opportunities & use of unique resources



Inclusive of low-tech business models via incubation and equity financing



Serving as model for other cities; flywheel of regional leadership

# The Oakland Equitable Climate Action Plan has several references relevant for the creation of a Green Bank and Green Opportunities Fund

|  GB  | Section                | Action                                                                                        | Relevance                                                                                                                                                                                                                                                                | Stakeholders                                                      |
|--------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  GOF | <b>City Leadership</b> | CL-4 Explore Creation of Public or Green Bank                                                 | Fossil fuel divestment; low-interest loans to member municipalities, funding for climate action activities                                                                                                                                                               | Finance; Public Bank East Bay advocates                           |
|                                                                                      | <b>Adaptation</b>      | A-1 Fund Creation and Operation of Resilience Hubs                                            | Community-serving facilities before/during/after natural hazard event                                                                                                                                                                                                    | Oakland Public Works (OPW), Office of Resilience (Resilience)     |
|                                                                                      |                        | A-2 Enhance Community Energy Resilience                                                       | Local renewable energy generation combined with energy storage; community-owned / shared solar                                                                                                                                                                           | OPW, EBCE/Ava                                                     |
|                                                                                      |                        | A-5 Identify and Reduce Financial Risks from Climate Change                                   | Risk mitigation incl. insurance products, green infrastructure bonds, real estate strategies                                                                                                                                                                             | OPW, Resilience                                                   |
|                                                                                      |                        | A-6 Expand and Protect Green Infrastructure & Biodiversity                                    | Installation and maintenance of projects and existing civic resources such as the parks system and public spaces                                                                                                                                                         | OPW, OES, Resilience                                              |
|                                                                                      | <b>Carbon Removal</b>  | CR-1 Develop Local Carbon Investment Program                                                  | “Carbon Neutral Oakland Business” designation; projects e.g., planting trees, replacing natural gas appliances, constructing EV charging stations, restoring creeks and parks; funding through offset / “Polluter Pays” fees / compliance credits                        | Office of Economic & Workforce Development Department (EWDD), OPW |
|                                                                                      |                        | CR-5 Assess Feasibility for Sequestration Incubator                                           | Carbon Sequestration Incubator to incubate and develop green jobs in urban agriculture, urban forestry, aquatic and riparian restoration, engineering technology, and/or other forms of carbon removal; while maximizing equity & avoiding harm to frontline communities | EWDD                                                              |
|                                                                                      |                        | CR-4 Explore Carbon Farming / CR-6 Explore Regional Aquatic Sequestration Opportunities       | Applying compost to soil on vacant public or private land; regional Bay Area coordination for aquatic approaches                                                                                                                                                         | PBD, OPW,                                                         |
|                                                                                      |                        | CR-2 Expand and Protect Tree Canopy Coverage; CR-3 Rehabilitate Riparian Areas and Open Space | Expanding and maintaining tree canopy coverage; restore creeks and provide ecosystem services                                                                                                                                                                            | OPW, Resilience                                                   |

## The Oakland Equitable Climate Action Plan has several references relevant for the creation of a Green Bank and Green Opportunities Fund



GOF

| Section                             | Action                                                                                                                                                                                                 | Relevance                                                                                                                                | Stakeholders                         |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Material Consumption + Waste</b> | MCW-3 Eliminate Single-Use Plastics and Prioritize Reuse in Food Preparation, Distribution, and Sale                                                                                                   | Supporting restaurants and developing a local supply chain                                                                               | OPW, StopWaste                       |
|                                     | MCW-4 Support the Reuse, Repair, Recovery, and Refurbishment Economy                                                                                                                                   | Supporting repair and recovery businesses; spaces dedicated to material repair and upcycling, and selling of repaired and upcycled goods | OPW                                  |
|                                     | MCW-5 Expand Community Repair Resources                                                                                                                                                                | Community sites and infrastructure for repair economy, including repair skills teaching and repair businesses                            | Oakland Public Library (OPL), EWDD   |
|                                     | MCW-2 Strengthen Infrastructure and Partnerships for Edible Food Recovery; MCW-6 Establish a Deconstruction Requirements                                                                               | Opportunities for food recovery, repair, and deconstruction businesses                                                                   | OPW, PBD                             |
| <b>Buildings</b>                    | B-2 Plan for All Existing Buildings to be Efficient and All-Electric by 2040                                                                                                                           | Financing energy efficiency and building envelope upgrades                                                                               | Planning & Building Department (PBD) |
| <b>Transportation + Land Use</b>    | TLU-3 Take Action to Reduce and Prevent Displacement of Residents and Businesses; TLU-4 Abundant, Affordable, and Accessible Public Transit; TLU-10 Expand Zero-Carbon, Shared-Use Bus and Van Service | Opportunities for low-carbon transportation business models                                                                              | DOT, PBD, EWDD                       |
| <b>Other priorities</b>             | 12. Support establishment of green, community- and worker-owned cooperative businesses to retain local wealth and build economic democracy                                                             | -                                                                                                                                        | -                                    |

 The IRA allocated \$27BN to mobilize clean financing and private capital, including funding green banks and similar organizations



## EPA Greenhouse Gas Reduction Fund

### \$14 billion National Clean Investment Fund

- 2-3 national clean financing institutions to partner with private sector to provide accessible, affordable financing for individuals, families, nonprofits, governments, small businesses, and others to deploy clean technology projects nationwide
- >40% of funds dedicated to low-income and disadvantaged communities

### \$6 billion Clean Communities Investment Accelerator

- 2-7 hub nonprofit organizations to provide funding and technical assistance to public, quasi-public, and non-profit community lenders, such as community development financial institutions, credit unions, green banks, housing finance agencies, and minority depository institutions
- 100% of funds dedicated to low-income and disadvantaged communities

### \$7 billion Solar for All competition

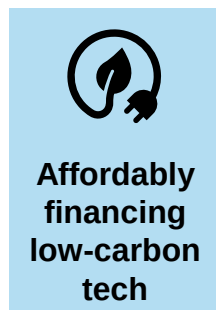
- Up to 60 grants to states, territories, Tribal governments, municipalities, and eligible nonprofits to create and expand low-income solar programs that provide financial and technical assistance, e.g., workforce development, to enable low-income and disadvantaged communities to deploy and benefit from residential solar

Deadline to apply: October 12, 2023  
Results: Spring 2024  
Potential applicants: Coalition for Green Capital

Deadline to apply: September 26, 2023  
Results: July 2024

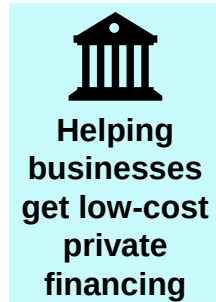
# California has informal Green Bank-like initiatives to both directly finance and reduce the risk of financing clean energy on homes and properties as well as small businesses

## Potential opportunities for a Green Bank in California



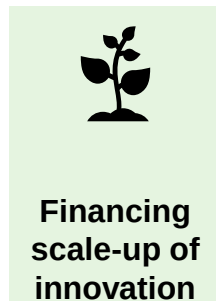
Property Assessed Clean Energy (PACE) financing secures a loan by linking payments to property taxes

Offering credit enhancements to financiers to kick-start private lending



Direct long-term, low-cost loans or credit enhancements for small businesses

Aggregating projects to improve access to low-cost private finance



Long-term loans, loan guarantees, insurance products, or low-cost public investment



**CA Green Bank:** Initiatives by State Treasurer's Office (STO) + California Infrastructure and Economic Development Bank (IBank)

Property Assessed Clean Energy Loss Reserve: mitigating the potential risk to lenders

GoGreen Home Energy Financing: facilitating attractive rates and terms on loans for approved energy retrofits by providing credit enhancement

Clean Diesel Truck Lending

Tax-Exempt, Private Activity Bonds

Small Business Credit Enhancements (CalCAP)

GoGreen Multifamily Energy Financing: credit enhancement to property owners for products incl. loans, leases, equipment financing agreements, service agreements and savings-based payment agreements

GoGreen Business Energy Financing: access for participating financiers to loss reserve fund that can cover up to 90% of losses

Sales and use tax (STE) exclusion to manufacturers that promote alternative energy and advanced transportation



A green bank can provide debt financing; equity financing through a separate fund would broaden the financing options for businesses

### Debt financing

Borrow funds that are repaid in full (principle) with interest

#### Examples

- Loans (e.g., bank loans)
- Venture debt (e.g., bridge/term loans, equipment financing)
- Bonds (e.g., corporate bonds)
- Secured debt (collateralized)
- Convertible loan note
- Project finance debt

#### Pros

- ✓ Non-dilutive: no loss of control and ownership
- ✓ Other instruments: (public green) bank can potentially provide other financial instruments such as tax credits, supply chain financing, credit enhancements / insurance

#### Cons

- × Default risk: failure to meet obligations can result in bankruptcy or loss of assets
- × Accessibility: Typically requires predictable cashflow or high interest for higher risk businesses / projects
- × Interest payments: Lenders have primarily financial obligations to recoup funds
- × Conditions: May be restricted by financial covenants, defined repayment terms, and other features to mitigate credit and other risks

### Equity financing

Funds in exchange for ownership stake (share of future profits + voting rights)

- Venture capital / private equity
- Warrants (e.g., SAFE)
- M&A
- IPO
- Crowdfunding
- Project equity

- ✓ No obligation to repay: Longer term capital where repayment is not contractually required, and interest payments don't tie-up cash flow
- ✓ Flexibility: Funds can be used for any purpose
- ✓ Business support: Investors may seek social and environmental returns in addition to financial returns and provide expertise and risk-sharing

- × Dilution: Give up decision-making control and upside (via dividend payments / capital gains)
- × Growth expectations: Investors may expect a certain speed and/or scale of growth



**Case study:** The Kapor Center focuses on intersection of racial justice and technology to create a more inclusive technology sector for all

## Kapor Capital

Investing in tech-driven early-stage companies committed to closing gaps of access in low-income and communities of color in US

### LPs

Fund III raised >\$120M incl. from Cambridge Associates, Align Impact, Ford Foundation, Bank of America, PayPal, Twilio, Footlocker, Blue Cross and Blue Shield of Louisiana, EIP Elevate Future Fund, Southern Poverty Law, The California Wellness Foundation, National Geographic Society, Winthrop Rockefeller Foundation and Stardust Equity

### Portfolio

- Blocpower: markets and finances clean energy and energy efficiency projects in financially underserved communities
- ChargerHelp: providing on-demand repairs and maintenance support from trained local workforces for EV charging stations
- Rain: early wildfire detection integrated with autonomous helicopters
- Aclima: environmental intelligence through sensor networks
- Compology: systems that monitors volume of waste inside dumpsters and then uses the data to route collection trucks more efficiently
- PlanetFWD: regenerative ingredient database

## Other activities

### Kapor Foundation

- Conducts research, make strategic grants, invest in entrepreneurs and funds, and inform equitable technology policy

### iLab Oakland

- 3-month residency for East Bay early-stage for or non-profit organizations seeking space and community to grow tech initiatives (as of 2019)



## Other initiatives and ecosystem in Oakland & Bay Area can be partners or inspiration for the Fund

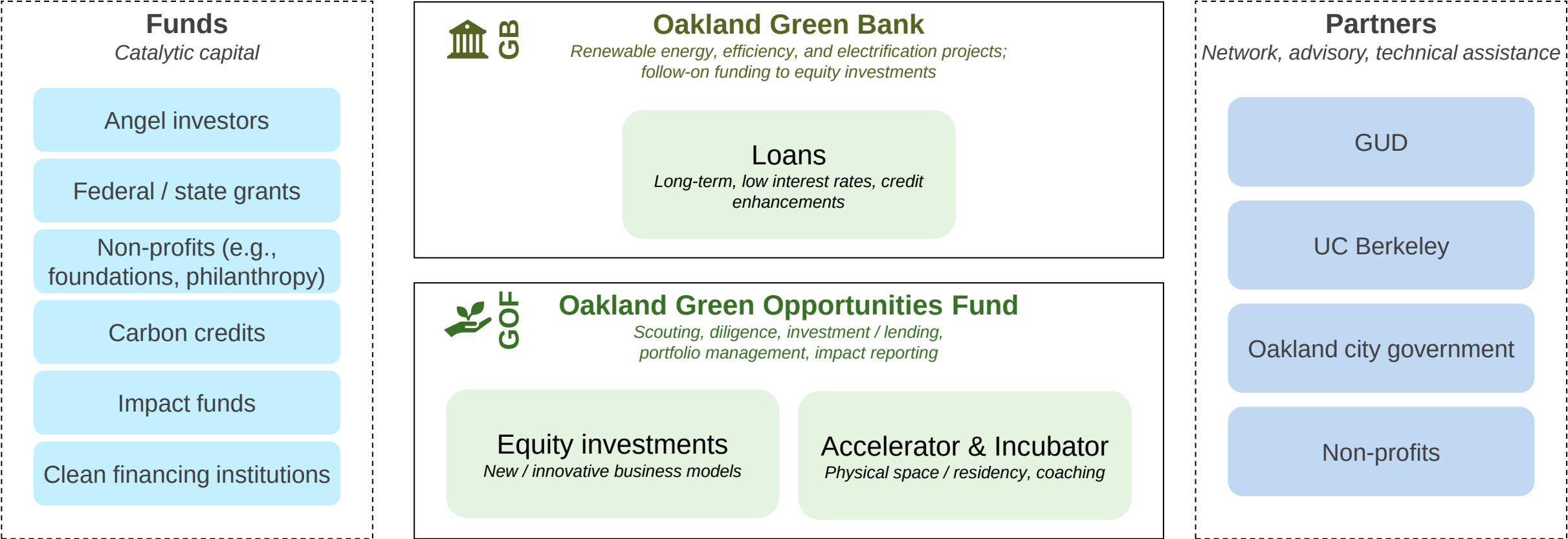
|            | Description                                                                                                                                                                                                                                                                                    | Funding structure                                                                                                                                                                 | Portfolio examples                                                                                                                                                             |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-profit | <br><u>GRID Alternatives</u><br>Nonprofit implementer of Clean Cars for All program and largest no-cost solar provider for low-income residents and workforce development                                     | Govt. grants, donor contributions, service contracts                                                                                                                              | N/A                                                                                                                                                                            |
|            | <br><u>Ava Community Energy</u><br>(fka EBCE)<br>Community choice aggregation program focusing on alternative energy, administrative and financial support for reducing emissions from buildings and vehicles | Operating revenues, grant income (e.g., California Arrearage Payment Program)                                                                                                     | Community innovation grants to non-profits and community orgs incl. Ecology Action, Community Impact LAB, West Oakland Environmental Indicators Project                        |
|            | <br><u>Yes SF Urban Sustainability Challenge</u><br>Challenge sourced innovative solutions to bring new life to downtown, buildings, and urban green and blue spaces                                          | UpLink (WEF), Deloitte, Salesforce, Citi, SF Chamber of Commerce, 20+ public and private sector partners commit in-kind and financial resources                                   | Taro.AI, Babylon, itselectric, Moxion                                                                                                                                          |
| For-profit | <br><u>Elemental Excelsior</u><br>Nonprofit investor in climate technologies with deep community impact, with cohort-based support                                                                            | Grants (\$32M from Office of Naval Research, part of \$44M Chan Zuckerberg Initiative)<br>Corporate members incl. Amazon, NextEra, Schneider Electric, United, Google, Fifth Wall | Capture6, Full Harvest, Nitricity, Bedrock Energy, Farm Link Hawai'i, Maui Nui Venison, Hawai'i 'Ulu Cooperative, Better Packaging Co., Allume Energy, Trove, WEEE Centre      |
|            | <br><u>Kapor Center</u><br>Focuses on intersection of racial justice and technology to create a more inclusive technology sector through research, community engagement, education, and VC                  | LPs, grants                                                                                                                                                                       | Blocpower, ChargerHelp, Rain, Aclima, PlanetFWD, Form Energy                                                                                                                   |
|            | <br><u>Powerhouse</u><br>Innovation firm & VC supporting entrepreneurs focused on solar and clean energy technologies                                                                                       | Corporate clients (e.g., Google, Enel, Mitsui); LPs (e.g., Credit Suisse, Microsoft, Total Energies, Toyota Ventures, Constellation)                                              | Presto (EV fleet charging); Helio (home electrification platform); Energetic Capital (unlocks project financing as managing general underwriter backed by reinsurance capital) |



## Initiatives and ecosystems in other regions also serve as a model and case study

|                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Funding structure                                                                                                                                                                                                                              | Portfolio examples                                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>State / municipal Green Banks</u></b> | Public, quasi-public or non-profit entity to facilitate private investment into domestic low-carbon, climate-resilient infrastructure; established at national, state (e.g., NY), county (e.g., Montgomery, MD), or city (e.g., DC) scales                                                                                                                                                                                                                                                                              | <u>DC</u> example: Funds consolidated into the Green Finance Authority Fund + additional funding sources e.g., District's Renewable Energy Development Fund (REDF) and Sustainable Energy Trust Fund (SETF) for commitments >\$100M over 5 yrs | DC example: Property Assessed Clean Energy (PACE); Small Business Loan Fund to upgrade to clean energy systems, increase stormwater resilience, and improve energy efficiency with <u>City First Enterprises</u> |
| <b><u>Halcyon</u></b>                       | Provides fellowship and/or investment for impact-driven businesses and entrepreneurs: <ul style="list-style-type: none"> <li>• Halcyon Fund: \$5 million fund to invest in pre-seed/seed Halcyon Fellows with avg. initial investment of \$100k</li> <li>• Halcyon Angels: network of angel investors in early-stage impact ventures</li> <li>• Microloan Fund: loans of up to \$25k to Halcyon ventures in DC area</li> <li>• Programs: Training, pro-bono services, residencies, mainly based in DC region</li> </ul> | Sponsors, donors, and partners e.g., Kimsey Foundation, Capital One, AWS, Deloitte, BoA, US EDA                                                                                                                                                | Solstice (acq by Mitsui MyPower Corp, enables access to community solar); Climatize (enables individuals to invest in solar projects); BoxPower (solar microgrid solutions)                                      |
| <b><u>Newlab</u></b>                        | Space and labs in NYC, Detroit, & Montevideo for deep tech startups and entrepreneurs, inventors, investors, working together to address critical challenges in energy, mobility, and materials                                                                                                                                                                                                                                                                                                                         | Partners for pilots, networking, and expertise: BHP, ConEdison, Crowley, Invest PR, NYC DOT, NYC EDC, Orsted, Verizon, MI Central, Ford, LG, Port of Monroe, Sorenson, Mercado Libre                                                           | Members: Amogy, Aclima, Gradient, VyCarb, Fermata Energy, itselectric                                                                                                                                            |
| <b><u>Emmerson Collective</u></b>           | Developing opportunities and solutions in education, environment, immigration, and health equity through philanthropy, creativity, and investments<br><u>Waverley Street Foundation</u> : nonprofit supporting farmers, students, Indigenous peoples, advocates advancing climate action and committed to spending endowment of \$3.5BN by 2035                                                                                                                                                                         | Laurene Powell Jobs                                                                                                                                                                                                                            | Emmerson: Aurora, Nithio, Twelve<br><br>Waverley: Grant partners incl Climate Imperative Foundation; National League of Cities; Just Transition Fund; Conservation International                                 |
| <b><u>CrossBoundary</u></b>                 | Advisory and investment firm, including: <ul style="list-style-type: none"> <li>• <u>Fund for Nature</u>: \$57M blended debt fund to finance projects in afforestation, reforestation, and revegetation (ARR) and reducing emissions from deforestation and forest degradation (REDD+) projects, starting in sub-Saharan Africa</li> <li>• CrossBoundary Energy (CBE): developer, owner, and operator of commercial and industrial renewable energy systems in Africa</li> </ul>                                        | Fund for Nature: SDG Impact Finance Initiative feasibility grant<br><br>CBE: Equity from KLP Norfund Investments AS                                                                                                                            | ---                                                                                                                                                                                                              |

An ecosystem of sustainable innovation financing in Oakland would support the creation and operation of the Green Bank and Green Opportunities fund



# The Green Bank and Green Opportunities Fund can support actions across the 'Four Greens'

Non-exhaustive

- Green Savings (ECAP Action MCW-5)**
-  **GB** - Lease financing / commercial mortgage for community site
  -  **GOF** - Temporary or permanent space for repair and skills training business to operate
  -  **GOF** - Financing for business models that help individuals / companies understand how to repair their goods and calculate savings

- Green Talent (ECAP Action MCW-4)**
-  **GB** - Equipment financing for repair tools and recovery equipment
  -  **GOF** - Temporary or permanent space for upcycling business to repair and sell goods
  -  **GOF** - Financing for business models that lower the cost or time for repair and upcycling
  -  **GOF** - Financing for business models that create innovative upcycled products

- Green Opportunities (ECAP Action MCW-3)**
-  **GB** - Loan for restaurant to fund upfront cost of changing practices
  -  **GB** - Loan for factory / manufacturer of circular materials, collateralized by equipment
  -  **GB** - Supply chain financing and tax credits / exclusions for manufacturing
  -  **GOF** - Financing for circular business models, e.g., reverse logistics providers, packaging providers

- Green Places (ECAP Action B-2)**
-  **GB** - Low-interest loans with different repayment structures and collateral, e.g., PACE
  -  **GOF** - Financing for business models that help train and/or provide workforce-on-demand for building upgrades
  -  **GOF** - Financing for business models that provide technologies that identify most cost-effective areas for upgrades
  -  **GOF** - Financing for business models developing technologies for the built environment

## Other resources

- [Silicon Valley Bank](#): Trends and insights on climate tech and startup financing
- [Green Bank Network](#): Resources and insights on green banks across the nation
- [Impact Assets 50](#): Database of impact investment fund managers
- [Project Drawdown](#): Resource on climate solutions
- [Canary Media](#): Clean energy journalism
- [Clean Energy Business Network](#): Resources on clean energy economy, including directory of clean energy technology & service providers